1.3 Transformations of Linear and Absolute Value Functions

Name _____ Period ____ Date _____

Algebra II 1.3 (30-minute plan) · Student sheet

I can…

- write functions that represent translations and reflections.
- write functions that represent stretches and shrinks.
- write functions that represent combinations of transformations.

Vocabulary · fill in the missing word

Horizontal translation — g(x) = _____

Vertical translation — g(x) = _____

Reflection in the x-axis / y-axis — g(x) = _____ = f(−x)

Vertical stretch or shrink — g(x) = _____

Horizontal stretch or shrink — g(x) = _____ when a > 1, a stretch when 0 < a < 1

Part 1 · Notes and examples (fill in each note, then work the example)

Inside or outside: Horizontal changes go inside: f(_____). Vertical changes go outside: _____.

Translations and reflections: Right h: f(_____) Up k: _____ Reflect in the y-axis: f(_____)

Example 1

f(x) = 2x − 1. Write a rule for g, a translation 3 units up, and for h, a translation 2 units left.

Example 2

f(x) = |x + 3| + 1. Write a rule for g, a reflection in the x-axis, and for h, a reflection in the y-axis.

Horizontal shrink: Horizontal shrink by 1/a: f(_____)

Example 3

a. f(x) = x + 2. Write g, a horizontal shrink by a factor of $\frac{1}{3}$.

b. f(x) = |x| − 1. Write h, a vertical stretch by a factor of 4.

Part 2 · Partner practice

Partner practice

$f(x) = 3x + 4$. Write g, a translation 5 units right.